Model Predictive Control Spring 2026
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Recitation 08

Exercise 1

Consider the system $x(k+1) = Ax(k) + Bu(k)$, $x \in \mathcal{X}$, $u \in \mathcal{U}$ with

$$\mathcal{X} = \{x \in \mathbb{R}^n | H_x x \leq b_x\}$$
$$\mathcal{U} = \{u \in \mathbb{R}^m | H_u u \leq b_u\}$$

and stage cost $l(x,u) = x^\top Q x + u^\top R u$ with $Q \in \mathbb{R}^{n\times n}, R \in \mathbb{R}^{m\times m}$ positive definite. Find a state-feedback $\kappa_f(x) = Kx, K \in \mathbb{R}^{m\times n}$, a corresponding terminal cost $V_f(x) = x^\top P x, P \in \mathbb{R}^{n\times n}$, and a terminal set $\mathcal{X}_f \neq \{0\}$ such that the following conditions are satisfied for all $x(k) \in \mathcal{X}_f$:

1. Terminal set is **invariant** under the local control law $\kappa_f(x(k))$:

$$x(k+1) = Ax(k) + B\kappa_f(x(k)) \in \mathcal{X}_f.$$

2. All state and input **constraints are satisfied** in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X},\ \kappa_f(x(k)) \in \mathcal{U}.$$

3. Terminal cost is a continuous **Lyapunov function** in the terminal set $\mathcal{X}_f$ and satisfies:

$$V_f(x(k+1)) - V_f(x(k)) \leq -l(x(k), \kappa_f(x(k))).$$

Exercise 2

Consider the system

$$\boldsymbol{x}(k+1) = \begin{bmatrix} 1.1 & 0.1 \\ 0 & 1 \end{bmatrix} \boldsymbol{x}(k) + \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} \boldsymbol{u}(k) \tag{1}$$

with initial condition $\boldsymbol{x}(0) = [-0.7, 1.4]^\top$, state constraints $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x}\|_\infty \leq 2\}$, input constraints $\mathcal{U} = \{\boldsymbol{u} \in \mathbb{R} : \|\boldsymbol{u}\|_\infty \leq 1\}$.

i) Write the constraints $\mathcal{X}$ and $\mathcal{U}$ in polytopic form, i.e., $\mathcal{X} = \left\{\boldsymbol{x} \in \mathbb{R}^2 | H_x \boldsymbol{x} \leq \boldsymbol{h}_x\right\}$ and $\mathcal{U} = \{\boldsymbol{u} \in \mathbb{R} | H_u \boldsymbol{u} \leq \boldsymbol{h}_u\}$.

ii) Define the system dynamics, constraints and initial condition in the function `generate_params`.

iii) In the class `MPC_DEMO`, implement the following model predictive controller using Yalmip[1]

[1] `https://yalmip.github.io/`, included in the Multi-Parametric Toolbox (MPT): `https://www.mpt3.org/`. See also the installation instructions in the project description of the MPC programming exercise.

$$
\begin{aligned}
\min_{X,U} \; & \sum_{i=0}^{N-1} \boldsymbol{x}_i^\top \boldsymbol{x}_i + \boldsymbol{u}_i^\top \boldsymbol{u}_i && && \text{(2a)} \\
\text{s.t.} \; & \boldsymbol{x}_0 = \boldsymbol{x}(k) && && \text{(2b)} \\
& \boldsymbol{x}_{i+1} = A\boldsymbol{x}_i + B\boldsymbol{u}_i, && i = 0, \dots, N-1 && \text{(2c)} \\
& H_x \boldsymbol{x}_i \leq \boldsymbol{h}_x, && i = 0, \dots, N-1 && \text{(2d)} \\
& H_u \boldsymbol{u}_i \leq \boldsymbol{h}_u, && i = 0, \dots, N-1 && \text{(2e)} \\
& \boldsymbol{x}_N = 0. && && \text{(2f)}
\end{aligned}
$$

Compare and discuss the closed-loop simulation of the controller *with and without* terminal constraint for different horizon lengths $N \in \{30, 15, 11, 10\}$.